North Carolina State University Invitational - Repaired practice edition - Answer key

Division B | Botany | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

4. (1 points) Authored replacement for source item 4: A supplied diagnostic report identifies Colletotrichum, a fungal genus, as the cause of a plant infection named anthracnose. Is this diagnosed infection fungal or bacterial? Explain using the report.

answer: Fungal: the supplied report identifies its causative organism as a fungus. This replaces the source question with more than one potentially fungal disease name. (1 points).

9. (1 points) Authored replacement for source item 9: Supplied crop survey: soybeans 40 hectares, corn 30, cotton 20. Which of these genetically modified crop plantings occupies the largest area in this survey?

answer: Soybeans, at 40 hectares. This is a supplied survey, not a claim about worldwide rankings. (1 points).

14. (1 points) Authored replacement for source item 14: A plant germinates, flowers, sets seed and dies within one growing season. Classify its life cycle as annual, biennial or perennial.

answer: Annual. The complete life cycle occurs in one growing season; no ambiguous climate-zone assumption is needed. (1 points).

21. (1 points) Authored replacement for source item 21: Compare supplied ages of individual trees: A=600 years, B=1,200 years, C=4,000 years. Which individual is oldest? Explain why this comparison does not establish the age of a clonal colony.

answer: Individual C, at 4,000 years. A single stem/tree age differs from the persistence of an entire clone; the given ages do not measure a colony. (1 points).

57. (1 points) Authored replacement for source item 57: The repair defines its reservoirs as sedimentary rocks, the atmosphere, living forest biomass and ice. Supplied carbon stores are 80, 10, 8 and 2 arbitrary units respectively. Which reservoir stores the most carbon in this model?

answer: Sedimentary rocks, at 80 units. This explicitly defined model avoids the source’s ambiguous use of biosphere. (1 points).

60. (1 points) Authored replacement for source item 60: An animal can digest starch but cannot hydrolyze the beta-1,4 glycosidic links in cellulose because it lacks cellulase. Explain why “it cannot break any glycosidic bonds” is too broad.

answer: It can hydrolyze other glycosidic linkages, including those in starch. The limitation is its lack of the appropriate cellulose-hydrolyzing enzyme, not an inability to break all glycosidic bonds. (1 points).

66. (1 points) Authored replacement for source item 66: A crop receives an inserted gene using a laboratory transformation method. Another crop is selected only through conventional crossing. Which is genetically engineered by the stated method?

answer: The crop receiving the inserted gene through laboratory transformation. This avoids the unverified earliest-commercial-crop assertion in the source. (1 points).